

AB RENEWABLE ENERGY ENGINEERING

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BIODIESEL METHYL ESTERIFICATION

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1. Product derived when methanol is removed from glycerol.

- a. Glycerine
- b. Methyl ester
- c. Glycol
- d. None of the above

2. Glycerol can be obtained at the ____ of a biodiesel plant.

- a. reaction tank
- b. methoxide tank
- c. settling tank
- d. All of the above

*biodiesel - top
glycerol - bottom*

3. Fatty acid methyl ester or mono-alkyl esters derived from vegetable oils or animal fats and other biomass-derived oils.

- a. Bioethanol
- b. Biodiesel
- c. Bio-oil
- d. All of the above
- e. None of the above

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4. Component of a biodiesel plant where potassium hydroxide and methanol are mixed together for treating the raw oil before undergoing transesterification process.

- a. Raw oil tank
- b. Reaction tank
- c. Methoxide tank
- d. All of the above
- e. None of the above

5. Heating value of biodiesel is approximately ____.

- a. 6,450 kcal/liter
- b. 9,000 kcal/liter
- c. 12,340 kcal/liter
- d. None of the above

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6. Vegetable oil can be used properly as fuel for compression-ignition engine when it is ____.

- a. mixed with diesel
- b. preheated
- c. mixed with gasoline
- d. None of the above

7. Which of the following is not a bio-chemical conversion of biomass?

- a. Bioethanol production
- b. Biogas generation
- c. Biodiesel production
- d. None of the above

8. Biodiesel is produced by subjecting ____ to potassium hydroxide and methanol mixture.

- a. coconut sap
- b. coconut water
- c. coconut oil
- d. None of the above

9. How much coconut oil is produced from one kilogram of copra?

- a. 40%
- b. 50%
- c. 60%
- d. None of the above

10. Process of producing biodiesel from coconut with the aid of heat element.

- a. Fermentation and distillation
- b. Transesterification
- c. Anaerobic digestion
- d. None of the above

11. Process of producing biodiesel from coconut oil without the need for heat element.

- a. Transesterification
- b. Cruzesterification
- c. Methyl esterification
- d. None of the above

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12. Average number of nuts a coconut tree can produce per month.

- a. 5 to 10 nuts
- ☒ b. 10 to 15 nuts
- c. 15 to 20 nuts
- d. None of the above

13. One liter of coconut oil requires ____ nuts in order to produce biodiesel.

- a. 5 to 6
- b. 7 to 9
- ☒ c. 10 to 11
- d. None of the above

14. Ideal moisture of jatropha seeds for the extraction of oil.

- ☒ a. 6%
- b. 8%
- c. 10%
- d. None of the above

15. Number of fruits that can be obtained per jathropa tree per 5 months.

- a. 540 fruits
- b. 780 fruits
- ☒ c. 960 fruits
- d. None of the above

760 fruits/jathropa/5 mos

16. Number of seeds produced per jathropa fruit.

- a. 1 seed
- b. 2 seeds
- ☒ c. 3 seeds
- d. None of the above

17. One liter of jatrophia oil can be produced from ____ of dried seeds.

- a. 2 kgs
- ☒ b. 4 kgs
- c. 6 kgs
- d. None of the above

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18. Product of mixing coconut oil with diesel to produce fuel suitable for compression-ignition engine.

- a. Biodiesel
- b. Bio-oil
- c. Bioethanol
- ☒ d. None of the above

19. Biodiesel in the settling tank can be found at the ____ of the container.

- ☒ a. Top
- b. Middle
- c. Bottom
- d. None of the above

20. Glycerol can be obtained from the biodiesel processing through ____ process.

- a. Fermentation
- b. Distillation
- ☒ c. Methyl esterification
- d. None of the above

21. Usual time required in mixing clean raw oil with KOH and methanol mixture (potassium methoxide solution) in the reaction tank.

- ☒ a. 1 hour
- b. 6 hours
- c. 24 hours
- d. None of the above

22. Recommended method of drying copra when use for biodiesel or coco methyl ester production.

- a. "Tapahan"
- ☒ b. Sundrying or air drying
- c. All of the above
- d. None of the above

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GOD BLESS !!!

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